

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent, in the Rocq prover. The first semantics closely models the ELPI’s implementation, it is based on a stack of choice points and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on and-or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics (1) to prove properties about the effect of the cut operator in the evaluation of choice points, since the rules from classical logic do not apply, and (2) to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [16, 17, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 22, 8, 9]. Other remarkable libraries include the Hierarchy-Builder library-structuring tool [5] and Derive [20, 21, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [10] to help users control backtracking. ROCQ users are familiar with functional programming, but not necessarily with logic programming; in particular, uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker allows the user to assert which predicates should be considered deterministic and then checks that these predicates behave like functions, i.e., they commit to their first solution and leave no *choice points* (places where backtracking could resume). A choice point correspond to an suspended *alternative* in the execution trace. In the following we use *choice point* and *alternative* as synonyms.

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [18] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations



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```

func f int -> int. % f is a deterministic predicate. The two rules
f 1 2.             % have non-verlapping heads, i.e. 1 != 2
f 2 3.
pred r int -> int. % r is a non-deterministic predicate. The heads
r 0 0.             % overlap: the query for 0 has three applicable
r 0 1.             % rules.
r 0 2 :- !.
func g int -> int. % g is deterministic: if the first rule succeeds, the
g A C :- r A B, f B C, !. % second one is cut away, as well as the choice
g D F :- f D E, f E F.   % points left by the call to r.

```

■ **Figure 1** Running example of a ELPI program

used by standard PROLOG abstract machines [23, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by cut are explicit and we prove the two semantics equivalent. Using the tree-based semantics, we then prove some properties about the impact of evaluating the cut in disjunctive queries. For example, unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 .

Finally, we use the formal semantics to prove the correctness of a static determinacy analysis [10]. Intuitively, nondeterminism arises when a computation can be completed by applying two different rules. We exclude this situation in two ways. Firstly, at most one rule should be used on a given query. This condition is satisfied if (i) the rules have non-overlapping heads: if two heads do not overlap in the inputs they accept, then no query can unify with both, or if (ii) we account for the presence of cut in rule premises: once a rule whose premises contain a cut succeeds, all remaining rules are discarded. Secondly, each rule of a deterministic predicate should (i) either call functions, this ensure that no choice point is left after the execution of the premises of the chosen rule, (ii) or call relations provided that they are followed by the cut operator, so that any allocated choice points preceding the cut are discarded.

We show that if every rule defining a deterministic predicate passes this analysis, then any call to that predicate leaves no choice points.

2 Preliminaries: syntax and informal semantics of cut

Both semantics share the same notion of program, written $\mathbb{P}$. Figure 1 is our running example. A program consists of a list of rules R together with a signature environment sigT . Besides the arity of each predicate, a signature specifies which arguments are inputs and which are outputs, and whether the predicate is deterministic. Inputs come before outputs.

A rule consists of a head term T_m and a list of premises. Each premise is an `Atom`, i.e. either the cut operator or a predicate call. Terms T_m include predicate symbols, data, and

```

Inductive Tm := ...
| Pred of P | Var of V.
Inductive sig := NonDet | Det.
Definition sigT :=
  P -> sig * nat * nat.

```

■ **Figure 2** Syntax of ELPI programs

```

Inductive A := cut | call : Tm -> A.
Record R := { head : Tm; premises : list A }.
Record P := { rules : list R; sig : sigT }.
Definition Σ := {fmap V -> Tm}.

```

■ **Figure 3** Syntax of ELPI programs

variables. In the concrete syntax, variables are written with capital letters, and predicate application follows the λ -calculus style (no parentheses).

The set of variables $\mathbf{V}$ is countable. We represent substitutions Σ as finitely supported maps from variables to terms, using the `finmap` library [15]. The empty substitution is written ε while the composition of two substitutions σ_1 and σ_2 with disjoint supports as $\sigma_1 + \sigma_2$.

The backchaining operation applies all rules to a query term; it is central to both semantics.

Definition `bc` : $\mathbb{P} \rightarrow \mathcal{F}_V \rightarrow \mathbf{Tm} \rightarrow \Sigma \rightarrow \mathcal{F}_V * \text{list } (\Sigma * \text{list } \mathbf{A}) :=$

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the backchain function takes a program, the set of variable names used so far ($\mathcal{F}_V$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term.

In our mechanization we abstract the unifier and its properties. The function `unify` takes two terms t_1 and t_2 and a substitution σ as input and returns an optional extension substitution σ' which is an extension of σ . The expected behavior of `unify` is the higher-order unification restrained into the pattern fragment as exemplified in [16]. Therefore if t_1 and t_2 unify under σ , then $\sigma' t_1 \equiv \sigma' t_2$.

2.1 The cut operator

The cut operator in ELPI implements the *hard cut* (see, e.g., the documentation of SWI-PROLOG [19]). Whenever the backchaining function `bc` returns a list of alternatives, the first one is explored immediately, while the others are suspended as *choice points*. Within a given alternative, premises are processed from left to right. When a cut is encountered, it discards (i) all choice points created by `bc` for the current call and (ii) all choice points created while executing the premises preceding the cut.

For example, consider the program P in Figure 1 and the query $\mathbf{g} \ 0 \ \mathbf{R}$. Both rules for `g` apply to this query. Backchaining therefore returns the following two alternatives:

$[(\sigma_1, [\mathbf{r} \ \mathbf{A} \ \mathbf{B}, \ \mathbf{f} \ \mathbf{B} \ \mathbf{C}, \ !]), (\sigma_2, [\mathbf{f} \ \mathbf{D} \ \mathbf{E}, \ \mathbf{f} \ \mathbf{E} \ \mathbf{F}])]$

with $\sigma_1 := \{\mathbf{A} \mapsto 0, \mathbf{C} \mapsto \mathbf{R}\}$ and $\sigma_2 := \{\mathbf{D} \mapsto 0, \mathbf{F} \mapsto \mathbf{R}\}$. For simplicity, we choose an example where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_V$ any further.

Execution continues with the first premise of the first alternative, namely `r A B` under σ_1 , i.e. `r 0 B`. The remaining alternatives are stored as choice points. Backchaining `r 0 B` returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [!])]$ where $\sigma_3 := \sigma_1 + \{\mathbf{B} \mapsto 0\}$; $\sigma_4 := \sigma_1 + \{\mathbf{B} \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{\mathbf{B} \mapsto 2\}$.

The next step of interpretation continues with `f B C` under σ_3 , that is, `f 0 R`, and leaves $[(\sigma_4, []), (\sigma_5, [!])]$ as new choice points. This time, backchaining fails (no rule for `f` matches input 0). We therefore backtrack to the most recently introduced choice point and retry `f B C` under σ_4 , i.e. `f 1 R`. This succeeds with $\sigma_6 := \sigma_4 + \{\mathbf{R} \mapsto 2\}$.

We now reach the cut. At this point there are still two unexplored alternatives: one from the third rule for `r` and one from the second rule for `g` (respectively, σ_5 and σ_2). Both are discarded, because they were created when, or after, the first rule for `g` was selected. (In contrast, a cut does not discard choice points created earlier; in this example, there are none.)

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Inductive tree :=
  | KO | OK                                     (* failure and success *)
  | Todo of Atom                               (* unexplored branch *)
  | Or of option tree &  $\Sigma$  & tree             (*  $A \vee B_\sigma$  *)
  | And of tree & list Atom & tree.           (*  $A \wedge_{B_0} B$  *)

Inductive tag := Expanded | CutBrothers | Failed | Success.
Definition step :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ 
Definition prune :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ 
Inductive runT :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop} :=$ 

```

■ **Figure 4** Signatures of the tree-based semantics

116 The final solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$.

117 In Sections 3 and 4, we provide fully mechanized operational semantics for PROLOG
 118 with cut. They differ in how they represent the ongoing computation, in particular how
 119 alternatives are stored and how they are pruned by cut.

120 *Notational conventions* In the following section we note $\mathbb{B}$ for the boolean type, and $\top$
 121 and $\perp$ for **true** and **false**. The option instances are noted $\bullet t$ for *Some* t and $\square$ for *None*.
 122 Following the Mathematical Components library, on which we depend, we use $x.1$ and $x.2$
 123 to denote the first and second projection applied to the pair x .

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 tion for list

3 And-Or Tree semantics

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An And-Or tree is a stratified, variable-width tree. It consists of two kinds of layers that alternate: a disjunctive layer is followed by a conjunctive layer and vice versa. At the root (a query), the tree records a list of alternatives (an Or-layer); for each alternative, it records a list of subqueries to be solved (an And-layer). Intuitively, solving a query requires solving one alternative in the Or-layer, but all subqueries in the corresponding And-layer.

The tree is built incrementally. The **Todo** node represents an unexplored branch (an **Atom**). When the atom is a **call**, we use the backchaining procedure from Section 2 to compute the two layers to attach to the tree: alternatives form the Or-layer, and the premises of each applicable rule form the corresponding And-layer. This expansion step is performed by the **step** procedure described in Section 3.2.

The Rocq data type for And-Or trees is given in Figure 4. In addition to **Todo**, it provides two distinguished nodes, **KO** and **OK**, which represent respectively the smallest failed and successful tree.

The **Or** node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing disjunction A . The left subtree A is optional and is absent once that branch has been fully explored. The right alternative is paired with a substitution σ , which becomes the current substitution when backtracking to B .

why option and
 not KO?

The **And** node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to be solved after A . We call B_0 the *reset point* for B : it is used to restore the initial state of B when backtracking occurs within A . An example is shown in Section 3.3.

A graphical representation of a tree is shown in Figure 6b. For compactness, we depict **And** and **Or** nodes as n -ary (rather than binary), with children associating to the right. The **KO** node acts as the terminating element of an **Or** list, while **OK** is the terminating element of an **And** list.

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Fixpoint next  $\sigma$   $t$  :  $\Sigma$  * tree :=
  match  $t$  with
  | Todo _ | KO | OK  $\Rightarrow$  ( $\sigma$ ,  $t$ )
  | Or  $\square$   $\sigma'$  B  $\Rightarrow$  next  $\sigma'$  B
  | Or  $\bullet$ A _ _  $\Rightarrow$  next  $\sigma$  A
  | And A _ B  $\Rightarrow$ 
    let ( $\sigma'$ , pA) := next  $\sigma$  A in
    if pA == OK then next  $\sigma'$  B
    else ( $\sigma'$ , pA)
  end.

Definition next_subst  $\sigma$   $t$  := (next  $\sigma$   $t$ ).1.
Definition next_tree  $t$  := (next  $\varepsilon$   $t$ ).2.

Definition success  $t$  := next_tree  $t$  == OK.
Definition failed  $t$  := next_tree  $t$  == KO.
Definition incomplete  $t$  :=
  if next_tree  $t$  is Todo _ then  $\top$  else  $\perp$ .

```

Figure 5 next and derived functions

3.1 The tree-based semantics

The tree is constructed incrementally by two recursive functions, **step** and **prune**. Both traverse the tree depth-first, left-to-right. The node to be explored in a tree is retrieved thanks to the **next_tree** (in Figure 5). In the snippet we also identify special kinds of trees: depending on the **next_tree**, we distinguish between *success*, *failed* and *incomplete* trees. Since all functions must terminate in Rocq, we define their transitive closure as the inductive relation **runT**.

Intuitively, **runT** iterates calls to **step** on incomplete trees to evaluate the **Todo** node. Upon failed trees, it calls **prune** to find the next alternative and continues from there, if one exists. We detail **step**, **prune**, and **runT** in Sections 3.2–3.4.

In Figure 6 we mark with a black arrow the result of **next_tree**.

3.2 The step procedure

The **step** procedure takes as input a program, a set of variables, a substitution, and a tree, and returns an updated set of variables, a **tag**, and an updated tree.

The set of variables is assumed to contain all variables occurring in the tree. It is passed to backchain so that rules can be refreshed correctly.

The **tag** (see Figure 4) indicates which kind of step was performed. **Failure** and **Success** are returned for trees that satisfy, respectively, the **failed** and **success** tests. **CutBrothers** indicates the interpretation of a superficial cut: **step** was called on an **And**-layer and its **next_tree** is the cut atom. Finally, **Expanded** accounts for the interpretation of predicate calls and non-superficial cuts.

The **step** procedure evaluates atoms. In particular, it leaves the tree unchanged when the tree is *success* or *failed*.

► **Lemma 1** (succ_step_iff). $\forall p \ v \ \sigma \ t, \text{ success } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Success}, t)$.

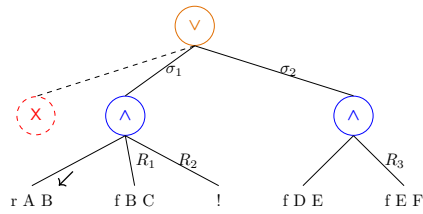
► **Lemma 2** (fail_step_iff). $\forall p \ v \ \sigma \ t, \text{ failed } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Failed}, t)$.

step expands **Todo** nodes by calling **backchain**, which returns a list l of alternatives. If l is empty, then the current call is replaced by **KO**. Otherwise, it is replaced by a right-skewed tree of **Or** nodes, where each leaf corresponds to a list of atoms $e \in l$. If e is empty, the leaf is **OK**; otherwise, it is a right-skewed tree of **And** nodes.

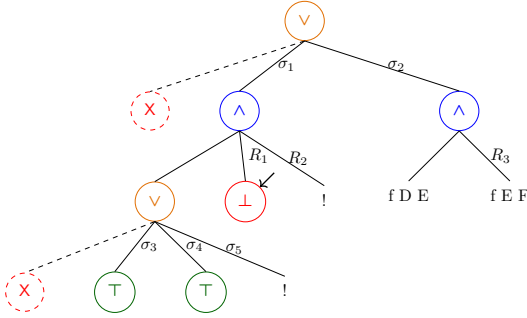
¹ The substitutions $\sigma_1 \dots \sigma_6$ are from Section 2.1. R_1, R_2 , and R_3 are reset points and correspond respectively to the lists of atoms $[f \ Y \ Z, !]$, $[!]$, and $[f \ Y \ Z]$.

g 0 R

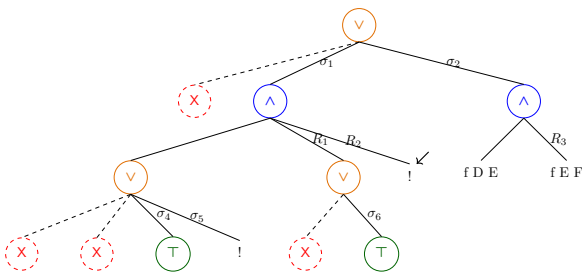
(a) Initial query



(b) Tree after step on g 0 R

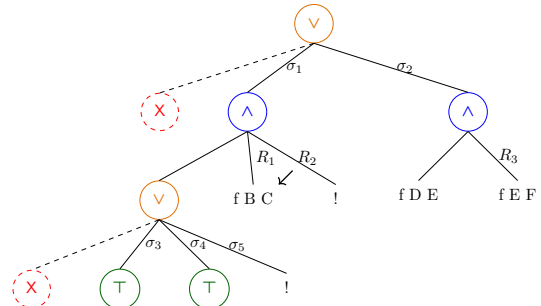


(d.1) step on Figure 6c

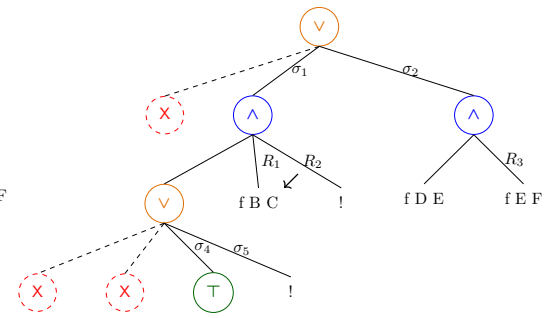


(e) step on Figure 6d.2

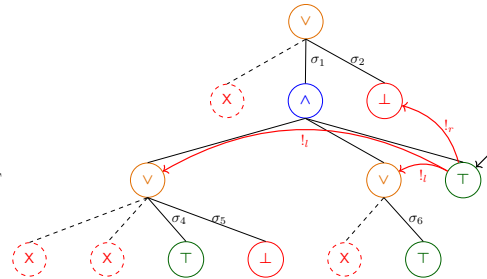
■ **Figure 6** Some tree representations¹



(c) step on Figure 6b



(d.2) prune on Figure 6d.1



(f) step on Figure 6e

Figure 6b depicts the tree corresponding to the running example (Section 2.1) as it undergoes calls to **step**. We run query $q \ 0 \ R$ against the program P in Figure 1. Let c_0 , c_1 , and c_2 be the children of the root. By construction, c_0 is the $\square$ tree, while c_1 and c_2 correspond respectively to the premises of rules r_1 and r_2 in P . We need the initial $\square$ subtree so that we can attach a substitution to c_1 . In particular, c_1 (resp. c_2) is associated with substitution σ_1 (resp. σ_2), whose values are the same as in Section 2.1. Reset points are marked with the R symbol: $R_1 := [f \ Y \ Z, !]$, $R_2 := [!]$, and $R_3 := f \ Y \ Z$. Intuitively, they record the lists of atoms used to generate the corresponding subtree. Other examples of **step** triggering backchaining appear in Figures 6c–6e.

sottolineare le
sostituzioni

3.2.1 The interpretation of a cut

The step corresponding to a cut performs two actions: (i) the cut node is replaced by OK; and (ii) the subtrees in the scope of **Cut** are pruned. More precisely, (ii) prunes the left siblings of the **Cut** node (in the same And-layer, denoted $!_l$) and prunes the right uncles of **Cut** (in the one-level-up Or-layer, denoted $!_r$).

An example of the impact of cut is shown in Figure 6f. The red arrows indicate the scope of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied to each pointed subtree. We note that the evaluation of a cut keeps the path leading to the cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under substitution σ_5 (which also succeeds) is replaced by KO. This amounts to discarding the third rule of predicate r (see Section 2.1).

3.3 The prune procedure

When backchain returns an empty list, **step** produces a *failed* tree and the computation must backtrack. The **prune** procedure is responsible for producing the new tree from which the computation can continue.

In Figure 6d.1 we encounter a failure because no rule applies to the atom $f \ B \ C$ under substitution σ_3 , which maps B to 0. The **prune** procedure prunes the path leading to this failure and resets the current sub-query to its original shape. More precisely, it kills the OK node under σ_3 (since that branch has been fully explored and produced no solution), and then replaces the KO node with the information stored in the reset point R_1 . This restores the call $f \ B \ C$, which can then be reconsidered under substitution σ_4 .

Furthermore, **prune** serves another important purpose. Its behavior depends on the boolean flag it receives as input: **prune** $\perp A$ recursively removes all failures (if any) from A , whereas **prune** $\top A$ removes the first success in A . For example, the tree in Figure 6f contains a successful path that maps R to 2. Calling **prune** on this tree returns $\square$, since there are no alternatives in the tree after the success..

Some interesting properties of **prune** are shown below; they illustrate how **prune** complements **step** with respect to failed, success, and **path_atom** trees.

► **Lemma 3** (*incomplete_prune_id*). $\forall b \ t, \text{incomplete } t \rightarrow \text{prune } b \ t = \bullet t$.

► **Lemma 4** (*prune_failedF*). $\forall b \ t \ t', \text{prune } b \ t = \bullet t' \rightarrow \text{failed } t' = \perp$.

► **Lemma 5** (*pruneFN_fail*). $\forall t, \text{prune } \perp t = \square \rightarrow \text{failed } t$.

3.4 The runT inductive

The inductive relation **runT** relates four inputs to three outputs. The inputs are a program, a set of variables, an initial substitution σ , and a tree t . The outputs are (i) an optional

221 result r of the form $\bullet(\sigma', t')$, (ii) a boolean b , and (iii) an updated set of variable names.

222 The main output is r : if $r = \square$, interpretation fails; otherwise σ' is the computed solution
 223 substitution and t' represents the remaining, not-yet-explored alternatives. The boolean b
 224 records whether a superficial cut was evaluated during execution, and the updated variable
 225 set tracks freshly generated names.

226 **runT** has four cases, depicted as inference rules in Figure 7. (STOPT) If **next_tree** t is a
 227 success, then the substitution σ' is extracted from t (starting from σ) and the input tree is
 228 cleaned of its successful path. (STEPT) If **next_tree** t is an atom, then **step** is invoked to
 229 evaluate t , and **runT** is called recursively on the updated tree. (BACKT) If **next_tree** t is a
 230 failure, then **prune** is called to clear the failed path; if the resulting cleaned tree exists, **runT**
 231 is called recursively on it. Finally, (FAILT) accounts for trees with no solutions.

232 These rules are deterministic:

233 ► **Lemma 6** (**runT_det**).

$$\forall p \ v_0 \ \sigma \ t \ r \ r' \ b \ b' \ v \ v', \text{ runT } p \ v_0 \ \sigma \ t \ r \ b \ v \rightarrow \\ \text{runT } p \ v_0 \ \sigma \ t \ r' \ b' \ v' \rightarrow r' = r \wedge v' = v \wedge b' = b.$$

234 The proof is by induction on the first **runT** derivation and is immediate, because the rules
 235 are selected based on **next_tree** t . In particular, the hypotheses *success* t , *path_atom* t ,
 236 and *failed* t are mutually exclusive. Moreover, by ??, the hypothesis of FAILT implies *failed*
 237 t ; hence FAILT is exclusive with STOPT and STEPT, and it is also exclusive with BACKT
 238 since they impose different requirements on the output of **prune**.

$$\frac{\text{success } t \quad \text{next_subst } \sigma \ t = \sigma' \quad \text{prune } \top \ t = t'}{\text{runT } p \ v \ \sigma \ t \ \bullet(\sigma', t') \perp v} \text{STOPT}$$

$$\frac{\text{incomplete } t \quad b' = (\text{tg} = \text{CutBrothers}) \vee b \quad \text{step } p \ v \ \sigma \ t = (v', \text{tg}, t') \quad \text{runT } p \ v' \ \sigma \ t' \ r \ b \ v''}{\text{runT } p \ v \ \sigma \ t \ r \ b' \ v''} \text{STEPT}$$

$$\frac{\text{failed } t \quad \text{prune } \perp \ t = \bullet t' \quad \text{runT } p \ v \ \sigma \ t' \ r \ n \ v'}{\text{runT } p \ v \ \sigma \ t \ r \ n \ v'} \text{BACKT}$$

$$\frac{\text{prune } \perp \ t = \square}{\text{runT } p \ v \ \sigma \ t \ \square \perp v} \text{FAILT}$$

■ **Figure 7** Rule system for **runT**

239 3.5 Properties of runT

240 Working with **runT** allows to prove some interesting properties about the behavior of the
 241 cut operator wrt disjunction, i.e. the **Or** node. We have proven several properties that are
 242 available at [this link](#). Below we show two of these lemmas.

243 ► **Lemma 7** (**runSST_or**).

$$\forall p \ v \ v' \ \sigma \ \sigma' \ l \ l' \ \sigma_1 \ r, \text{ runT } p \ v \ \sigma \ l \ \bullet(\sigma', \bullet l') \top v' \rightarrow \\ \text{runT } p \ v \ \sigma \ (\text{Or } \bullet l \ \sigma_1 \ r) \ \bullet(\sigma', \bullet(\bullet l' \vee \text{KO}_{\sigma_1})) \perp v'.$$

244 ► **Lemma 8** (**runNT_or**).

$$\forall p \ v \ v' \ \sigma \ t \ \sigma_1 \ t', \text{ runT } p \ v \ \sigma \ t \ \square \top \ v' \rightarrow \\ \text{runT } p \ v \ \sigma \ (\bullet t \vee t'_{\sigma_1}) \ \square \perp \ v'.$$

245 ► **Lemma 9** (run_orSST).

$$\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ l \ l' \ r \ r', \text{ runT } p \ v \ \sigma \ (\bullet l \vee r_{\sigma_1}) \bullet(\sigma', \bullet(\bullet l' \vee r'_{\sigma_1})) \perp \ v' \rightarrow \\ \exists b, \text{ runT } p \ v \ \sigma \ l \bullet(\sigma', \bullet l') \ b \ v' \wedge r' = \text{if } b \text{ then } \text{KO} \text{ else } r.$$

246 In ??, we evaluate the tree l . This evaluation produces a success with signature σ' and
 247 a new tree l' . During the evaluation, a superficial cut in l is traversed. This implies that
 248 the evaluation of $\bullet l \vee_{\sigma_1} r$ returns σ' as substitution, and the alternatives are $\bullet l' \vee_{\sigma_1} \text{KO}$. The
 249 right-hand side of the Or node is replaced with KO by $!_r$.

250 In ??, we are in a similar scenario: t evaluates a superficial cut but, this time, produces a
 251 failure. This implies that the evaluation of $\bullet t \vee_{\sigma_1} t'$ also fails.

252 Finally, in ??, we evaluate $\bullet l \vee_{\sigma_1} r$. The evaluation succeeds, producing the result
 253 $\bullet(s_1, \bullet l' \vee_{\sigma_1} r')$. This means that l still has l' as unexplored alternatives. We deduce that
 254 the evaluation of l produces the substitution σ_1 with alternatives l' . However, we have no
 255 information about whether a superficial cut has been crossed in l . If this is the case, we
 256 know that r' is cut away by $!_r$; otherwise, r' has not been explored and is equal to r .

257 From these lemmas, it is clear how the order in which alternatives are evaluated changes
 258 the behavior of the program.

259 4 Stack semantics

260 We now introduce the stack semantics. The interpreter is close to that of the ELPI language
 261 and provides an operational semantics that is similar to the one proposed by Pusch in [18],
 262 which is in turn closely related to the semantics given by Debray and Mishra in [6, Section 4.3].
 263 Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that
 264 are also present in the Warren Abstract Machine [23, 1].

265 We represent a state of the ELPI language thanks to the two mutual inductives shown
 266 below.

```
Inductive alts := nilA | consA of (Σ * goals) & alts
with goals := nilG | consG of (A * alts) & goals .
```

267 A stack is an enhanced two-dimensional list, called *alts* in the inductive. The outermost
 268 list represents a list of alternatives in disjunction, each accompanied by the substitution that
 269 should be used for their interpretation. The innermost list is a list of atoms, that is, the
 270 list of goals in conjunction. These goals are decorated with a pointer to an alternative list,
 271 which is used to keep track of where to jump when a cut is interpreted. We call these special
 272 alternatives the *cut-to* alternatives.

273 The stack interpreter is called **runS** and has the following type.

$$274 \text{ Inductive runS } (p: \mathbb{P}) : \mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop} := \quad (1)$$

275 The inductive **runS** represents a routine taking as input a set of variables and a list
 276 of alternatives and returning an option. $\square$ stands for interpretation with no solution,
 277 whereas $\bullet(\sigma, a)$ represent a successful interpretation with σ as solution and a as the list of
 278 non-yet-explored choice points.

279 The rule system for **runS** is given in Figure 8 and is made by 4 cases. We distinguish
 280 two main cases: if the list of alternatives is empty, (FAILE) we have a failure: we stop the
 281 computation and return $\square$. If the list of alternatives is the list $(\sigma_0, h) :: a$, the interpreter

se riesco success
 (A or B) = suc-
 cess (B or A) if
 no sup cut in A
 and B

Dire che non ab-
 biamo le due
 info ritornate da
 runT

Definition `save_gs` $a \ g \ b :=$
 $[\text{seq } (x, a) \mid x <- b] ++ g.$

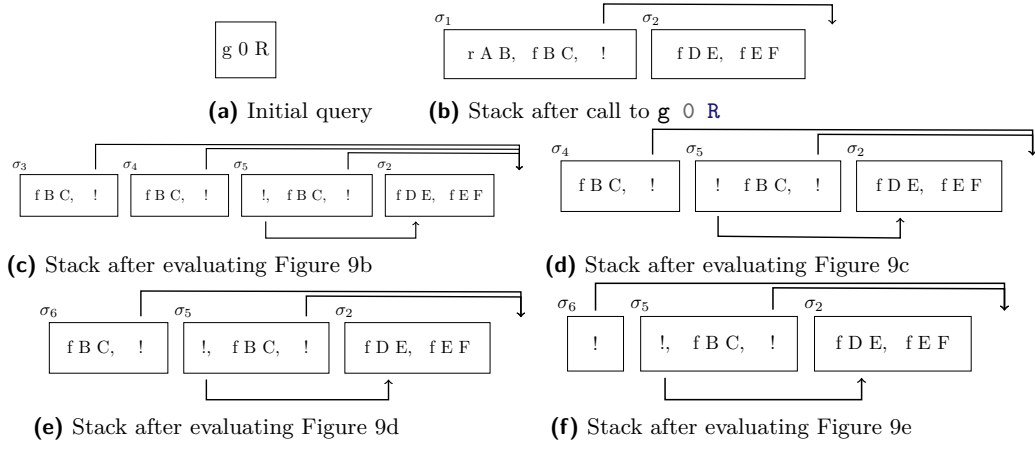
Definition `save_as` $a \ g \ b :=$
 $[\text{seq } (x.1, \text{save_gs } a \ g \ x.2) \mid x <- b].$

Definition `stepS` $p \ v \ t \ \sigma \ a \ g :=$
 $\text{let } (v', r) := \text{bc } p \ v \ t \ \sigma \ \text{in}$
 $(v', \text{save_as } a \ g \ r).$

$$\frac{}{\text{runS } p \ v \ ((\sigma, \emptyset) :: a) \bullet (\sigma, a) \text{ STOPS}} \quad \frac{\text{runS } p \ v \ ((\sigma, g) :: ca) \ r}{\text{runS } p \ v \ ((\sigma, (\text{cut}, ca) :: g) :: _) \ r} \text{ CUTS}$$

$$\frac{\text{stepS } p \ v \ t \ \sigma \ a \ g = (v_1, a') \quad \text{runS } p \ v_1 \ (a' ++ a) \ r}{\text{runS } p \ v \ ((\sigma, (\text{call } t, _) :: g) :: a) \ r} \text{ CALLS} \quad \frac{}{\text{runS } p \ _ \ \emptyset \ \square \text{ FAILS}}$$

■ **Figure 8** Rule system for `runS`



■ **Figure 9** Example of `runS` execution

282 evaluates the first choice point h under the substitution σ_0 . If h is empty, (STOPE), then we
 283 have a success, i.e. we return the input σ_0 and leave a as the future choice points. Otherwise,
 284 h is the list $(t, ca) :: g$ where t is the first atom to evaluate, ca is its cut-alternative list and g
 285 is the list of future conjuncts. If t is a cut (CUTE), we need to keep evaluating g under σ_0
 286 but we change the alternatives to ca . Finally, in the case of a call (CALLE), we backchain
 287 the call in the program. If rs is the list of rule premises returned by `bc`, then each list of
 288 premises if mapped to a list of goals (`goals`) have a as cut-alternative and the global list is
 translated to a list of alternative (`alts`).

Dire una parola
 su backtracking
 = tl if bc = nil

292 *Example of execution* We propose an example of the execution of `runS` in Figure 9. The
 293 arrows in the images represent the cut-alternative pointers of the cut. We have not drawn
 294 arrows coming out of atom-calls since the interpreter ignores the cut-alternatives in this case
 295 (cf. CALLE). The evolution of the stack in Figure 9 mirrors the example in Section 2.1: we
 296 evaluate the first goal of the first alternative. During backchaining, we add a pointer to
 297 the remaining alternatives to each new cut operator. If a cut is evaluated, we replace the
 298 remaining alternatives with the cut-to alternatives. For example, in Figure 9f, evaluating
 the cut discards the second and third elements in the stack. Backtracking is performed by
 simply removing the first alternative from the stack, as shown in Figure 9d.

299 *Remark* We want finally to point out how the lemma that we were able to prove in
 300 Section 3.5 are not easy to be expressed in the stack semantics. This is because there is
 301 no sufficient structure in the stack to understand what is the scope of the cut operator.

For example, in a stack made of these alternatives $a_0, a_1 \dots a_n$, we have no clear idea of what happens if the first alternative a_0 succeeds: there could be a cut in a_0 but, is this cut superficial? What is its impact wrt the alternatives $a_1 \dots a_n$? More then that, it is also possible that the state is hill-formed and the cut does not point to a suffix in the list of alternatives.

5 Equivalence of the two semantics

In order to relate the two semantics we have to relate the computations states, i.e., the And-Or tree and the stack of alternatives. In particular we define a translation from trees to stacks, called `t2l`. We do not explicitly provide the converse translation, an economy allowed by the proof technique we employ, inspired by [2].

Before describing the translation of trees to stacks (in Section 5.2) we need to define the notion of *valid tree*. The `tree` datatype indeed contains trees that cannot be generated by `runT` via `step` or `prune`, for example all the trees that do not correspond to a depth-first left-to-right tree construction strategy.

5.1 Valid trees (`valid_tree`)

The class of valid trees is characterized by the function shown in Figure 10a.

While all base cases of tree are considered valid, we need to analyze carefully the cases for the `Or` and `And` constructors.

For the `Or` constructor, we distinguish two cases depending on whether the left hand side is present. If it is not present, then the right hand side must be a valid tree. Otherwise, the the left hand side must be a valid tree and the right hand side must either be the `K0` tree or be unexplored. The first case occurs when the right hand side is cut out by the evaluation of a superficial cut present in the left hand side. In the latter case we use the `base_or` predicate to check that `B` is the right-skewed tree formed by a disjunction of conjunctions that is generated after a succesful backchain to a `call`.

For the `And` constructor, the left hand side is required to be a valid tree. The shape of the right hand side depends on whether the left hand side is a success. If it is not successful, then the right hand side must not be explored, i.e. `B` must be the right-skewed tree containing conjunctions of premises atoms. This is enforced using the `big_and` produce that generates a tree out of the reset point B_0 , the same procude used to transform each alternative output by `bc` into the And-layer of the resulting tree. When the left hand side is successfule, then the right hand side may have been explored, and consequently must be a valid tree.

5.2 From trees to stacks (`t2l`)

The `t2l` recursive function translates a tree to a stack. For space constraints Figure 10b only gives the base cases, while we describe the recursive cases in the following text. DAVVERO?

The function `t2l` takes the tree t_0 , a substitution σ_0 and a list of choice points cp . These choice points represent alternatives that appears at the same level of the current tree, such as, for example, the right-siblings of a `Or` node. The substitution TODO

The `OK` node represent one succesful alternative, thereofre it is translated into a singleton list with the empty list of goals. The `K0` node represent a failure, i.e. it is mapped to the empty list of alternatives. Finally, atoms are mapped to one alternative containing a single goal with the empty cut-to alternative list. They cannot point to cp since they are not right-uncle subtrees, i.e. EXPLAIN. TODO

```

Fixpoint valid_tree A :=
  match A with
  | Todo _ | OK | KO => T
  | Or □ _ B => valid_tree B
  | Or •A _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

Fixpoint t2l t σ (cp: alts) : alts :=
  match t with
  | OK => [(σ, ∅)]
  | KO => ∅
  | TA a => [(σ, [(a, ∅)])]
  ...

```

(a) Valid tree

(b) Tree to list

■ **Figure 10** Valid tree and Tree to list

345 The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for
 346 B . The alternatives of B are valid choice points for A and should be passed to the translation
 347 of A . The two list of alternatives can be concatenated and, each atom, should add cp as
 348 new cut-to alternatives: A and B are one level lower wrt cp and therefore the cut they have
 349 should point to cp .

350 The translation of $A \wedge_{B_0} B$ starts with the translation of A . If the translation of A
 351 is an empty list we return the empty list of alternatives without even touching B nor B_0 ,
 352 since an empty translation for A indicates failure and this in turn implies the failure of the
 353 entire conjunction. When the translation of A is a list $(\sigma_0, g) :: a$ the B node represent the
 354 continuation of the first alternative g , whereas B_0 is the continuation of each alternative in a .

superficial?

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all'inizio

355 5.2.1 Example of t2l

356 The translation of the trees in Figure 6 is given in Figure 9. The letters in the figures
 357 relate the trees to the corresponding letters in the stack figures. For example, the tree 6b is
 358 translated into 9b. We note that the two trees 6d.1 and 6d.2 are both translated into 9d,
 359 showing that $t2l$ is not injective; that is, there are different trees that map to the same stack.
 360 This means that there are transitions in runT that are not visible in runS .

361 As a more precise example of the behavior of $t2l$, we consider the tree 6c, and in
 362 particular its deepest Or node. This subtree shows a disjunction with four children. The
 363 first is ignored, since it is $\square$. The success node below σ_3 has $\mathbf{f} \ B \ C$ and the cut operator
 364 as continuation, corresponding to the first cell in the stack in 9c. We note that the cut
 365 operator points outside the stack, since the scope of this cut eliminates its right uncle. The
 366 success node below σ_4 has R_1 as continuation, where R_1 is the list of goals $\mathbf{f} \ B \ C, !$; this is
 367 translated into the second cell of the stack in 9c. Finally, the cut operator under σ_5 also has
 368 R_1 as continuation. It corresponds to the third alternative in Figure 9c. We can see that the
 369 first cut points to the translation of the subtree under σ_2 , since it is one OR-layer above its
 370 right uncles.

371 5.3 Equivalence proof

372 The tree-based semantics exposes more information so to be usable to express the lemmas in
 373 Section 3.5. For the equivalence this information is irrelevant, hence define a version of runT
 374 that hides it.

375 **Definition** $\text{runT}' \ p \ v \ \sigma \ t \ r := \exists \ b \ v', \text{runT} \ p \ v \ \sigma \ t \ r \ b \ v'.$ (2)

From ??, we observe that, upon backtracking, `runT` may perform more reduction steps than `runS` before returning a solution or a failure. These transitions are called silent transitions and must be skipped when proving the equivalence between the two semantics. Therefore, we state the following lemma.

► **Lemma 10** (`pruneF_t2l`).

$$\begin{aligned} \forall t t' b, \text{valid_tree } t \rightarrow \text{failed } t \rightarrow \text{prune } b \ t = \bullet t' \rightarrow \\ \forall l \sigma, \text{t2l } t \ \sigma \ l = \text{t2l } t' \ \sigma \ l. \end{aligned}$$

Proof. By induction on the tree t .

◀ dire di v

The first direction of the equivalence states that the execution of a valid tree is matched by the execution of the stack obtain by translating the tree and moreover the unexplored alternatives returned as a tree match the ones returned as a stack.

► **Theorem 11** (`tree_to_elpi`).

$$\begin{aligned} \forall p \ t \ \sigma \ r, \text{let } v := \text{vars_tree } t \ \backslash \ \text{vars_sigma } \sigma \text{ in} \\ \text{valid_tree } t \rightarrow \\ \text{runT}' \ p \ v \ \sigma \ t \ r \rightarrow \\ \text{let } a := \text{t2l } t \ \sigma \ \emptyset \text{ in} \\ \text{let } r' := \text{if } r \text{ is } \bullet(\sigma', t) \text{ then } \bullet(\sigma', \text{t2l } (\text{odflt } KO \ t) \ \sigma \ \emptyset) \\ \text{else } \square \text{ in} \\ \text{runS } p \ v \ a \ r'. \end{aligned}$$

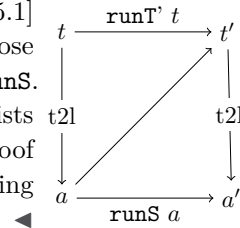
Proof. We proceed by induction on the execution of `runT'`. There are four cases to consider. The first case arises from `STOPT`; it deals with successful trees. A successful tree must start with an empty list, and therefore we conclude using `STOPE`. The case for `STEPT` requires treating two subcases: `step` evaluates either a cut or a call. Accordingly, we apply `CUTE` or `CALLE`, followed by the induction hypothesis. The case for `BACKT` is silent: it represents the non-injective behavior of `t2l`; however, thanks to Lemma 10 and the induction hypothesis, the conclusion is proved. The last case arises from the derivation `FAILT`. It is easily proved using `FAILE` and the fact that if `prune` returns $\square$ when trying to erase failure in t , then `t2l` applied to t returns the empty list.

The converse direction is stated following [2]: instead of using a function to translate a stack into a tree it states that any tree that translates to the given stack has an equivalent execution, i.e. gives the same solution and returns a unexplored tree that translates to the unexplored stack.

► **Theorem 12** (`elpi_to_tree`).

$$\begin{aligned} \forall p \ v \ a \ r, \text{runS } p \ v \ a \ r \rightarrow \\ \forall \sigma \ t, \text{valid_tree } t \rightarrow \text{t2l } t \ \sigma \ \emptyset = a \rightarrow \\ \text{if } r \text{ is } \bullet(\sigma', a') \text{ then} \\ \quad \exists t', \text{runT}' \ p \ v \ \sigma \ t \bullet(\sigma', t') \wedge \text{t2l } (\text{odflt } KO \ t') \ \sigma \ \emptyset = a' \\ \text{else } \text{runT}' \ p \ v \ \sigma \ t \ \square. \end{aligned}$$

Proof. The proof is based on the idea explained in [2, Section 3.5.1] and depicted in the figure on the right: we have a state t whose translation is a , we proceed by induction on the execution of `runS`. This gives us not only the output a' but that hypothesis that it exists a tree t' such that its translation to the stack state is a' . This proof strategy is indirectly providing a kind of `l2t` function by combining the execution of the `runS` and `t2l`.



401 Stating the converse direction in a symmetric way would have requires a function 12t
 402 transforming an stack a to a tree t . Writing this function is far from trivial, since the
 403 structure of the tree is lost by *save_alts* and run that intuitively keep the state into a big
 404 disjunction of conjunctions by distributing conjunctions over disjunctions. Moreover one
 405 would need to define a notion of valid stack that is equally intricate.

406 6 Case study: determinacy alanysis

407 An interesting application of the tree semantics is the determinacy checking that was added
 408 in version 3 of the ELPI language [10]. The checker is meant to receive the signature of the
 409 predicates declared by the user and then verify that the rules of a deterministic predicate
 410 will return at most one solution, we define this particular kind of predicate as functions.
 411 Thanks to this static verification, the user can better control unexpected backtracking at
 412 compile time: if an incoherence is rilevata then a fatal error explain why the current program
 413 may break determinacy. A function behaves deterministically if two main conditions are
 414 respected: *mutual exclusion* guarantees that at most one rule can be successfully used on a
 415 given query, whereas *rule checking* ensure that each rule does not creates choice points, e.g.
 416 by calling non-deterministic predicates.

417 The syntax of the signatures in Figure 1 indicates that **f** and **g** are expected to be
 418 functions, as indicated by the **func** keyword. The **->** symbol separates inputs from outputs;
 419 therefore, all these predicates are expected to take an **int** as input and return an **int** as
 420 output.

421 The signatures of the program are stored in its **sig** projection of the program P passed to
 422 the interpreter, and are encoded as described in [10], tahs is, similar to the code in Figure 1,
 423 we have arrows decorated with an input mode for term application, base types (to encode, for
 424 example, *int*), and two special symbols to distinguish deterministic from non-deterministic
 425 predicates.

426 A program execution behaves deterministically if given a program, a substitution and a
 427 tree to the **runT** inductive, then either the execution fails, or, if it succeeds, then the tree of
 428 alternatives is $\square$. Below we define **is_det**.

429 **Definition** **is_det** $p \sigma v t := \forall r, \text{runT}' p v \sigma t r \rightarrow r = \square \vee \exists \sigma, r = \bullet(\sigma, \square)$.

use coq syntax instead of elpi? The theorem we want to prove is the following.

431 ► **Theorem 13** (**det_check_call**).

$$\forall p \sigma t v, \text{acyclic_sigma } \sigma \rightarrow \text{check_}\mathbb{P} p \rightarrow \text{tm_is_det } p.(sig) t \rightarrow \text{is_det } p \sigma v \text{ (Todo (call t))}.$$

432 With **check_** $\mathbb{P}$ being the function checking the program wrt mutual exclusion and rule
 433 checking and **tm_is_det** being the function testing if the term head refers to a rigid constant
 434 with deterministically type. The proof of this lemma should be done on the derivation of the
 435 program execution. However, in this definition, the induction hypothesis is too weak, since
 spiega notazioni in we would have to prove ... but we will not be able to show that the ... is deterministic.

437 To solve the problem, we need to work on “deterministic trees” (**det_tree**), show that if
 438 a tree respect this condition, and then prove the following theorem which is more generic
 439 then **det_check_call**.

440 ► **Lemma 14** (**det_check_tree**).

$$\forall \sigma v p t, \text{acyclic_sigma } \sigma \rightarrow \text{acyclic_sigmaT } t \rightarrow \text{check_}\mathbb{P} p \rightarrow \text{det_tree } p.(sig) t \rightarrow \text{is_det } p \sigma v t.$$

Since `det_check_call` is an instance of `det_check_tree`, its proof is now trivial.

As an important remark, proving the same lemma with the stack semantics, it is difficult to correctly define the predicate `det_stack`, checking for the deterministic behavior of a generic stack, since, similar to Section 3.5, the lack of structure make this goal hard. We need therefore an intermediate data structure, like the tree to encompass this problem.

7 Related work

We studied a PROLOG semantics in which input arguments are *matched*, rather than unified, against rule heads. This choice agrees with the ELPI interpreter, which is in turn inspired by B-Prolog's "matching-clauses" [24]. The two semantics we presented (and proved equivalent) also apply to standard PROLOG if one forces all predicate signatures to have only output arguments.

For determinacy analysis, matching input arguments affects the development only as follows: axiom XXX does not require the query q to be ground. Adapting our results to standard PROLOG therefore requires two changes. First, to add the corresponding groundness hypothesis to XXX. Second, to discharge this premise when XXX is used. Typically this requires an additional static analysis that checks whether the program is well-moded. Well-moded predicates produce ground outputs assuming ground inputs; hence, starting from a ground initial query, well-moded programs behave exactly as if input arguments were matched, and hence the proofs in Section 6 still apply.

The only mechanization of Prolog's semantics we could find is the one by Pusch in ISABELLE/HOL [18] that is based on the model of Debray and Mishra [?, Sec. 4.3]. Their work start from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [23, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in Section 4 that pushed us to develop an more explicit semantics (with respect to cut). In some way our work is complementary, showing the stack based semantics equivalent to a more optimized one.

Another relevant work are the ones by King [14], but we could not find their Rocq source code. Their semantics is denotational and cannot easily cover all the features our paper [10] does, but would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by [2] and we thank Y.Bertot for insightful discussions on the topic. In [2] Bertot relates bla bla, witout never constructing a decompiler, exactly as we never build a list to tree function.

FINIRE

8 Conclusion

We do this. This is a first step for XXXX. We need to add substitutions. The missing part of the proof becomes related to abstract interpretation, since the det types for a lattice and one has to prove that the concrete runtime values variables take agree with the abstraction computed statically. This proof is ongoing and somehow orthogonal to the current one since the cut operator and the HO feature do not interfere with eachother in complex ways.

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